

Electroresponsive Weak Polyelectrolyte Brushes

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Cite This: *Macromolecules* 2022, 55, 2636–2648



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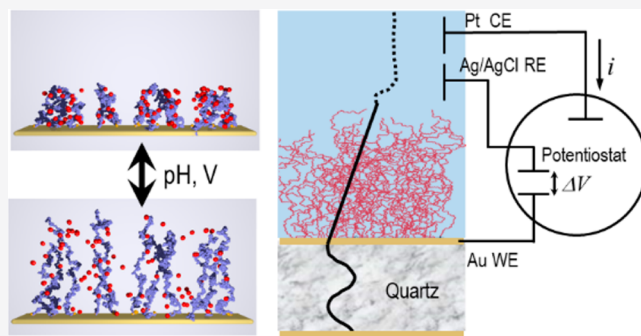


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Supporting Information

ABSTRACT: We explored the effect of an applied electric field on the molecular conformation of two weak polyelectrolytes grafted on gold electrodes—poly(acrylic acid) and the weak polybase poly(2-(dimethylamino)ethyl methacrylate)—by combining a quartz crystal microbalance with an electrochemical setup. Our results indicate that different mechanisms influence the response of weak polyelectrolytes to an externally applied potential difference, including electrochemical modification of the local pH by oxygen reduction and the electrostatic interaction between the metallic electrode and the charged monomers and ions in solution. Our results evidence that fine control of polymer conformation can be readily obtained at low applied potentials, opening pathways for precise, localized control of surface properties. However, these applications require considering the history-dependent behavior of polyelectrolyte brushes, hysteresis in the response, and the influence of electrochemical reactions driven by potential changes.



INTRODUCTION

Active regulation of surface properties like wettability, adhesion, or friction opens exciting possibilities in many different fields, from microfluidics to water recovery. One widespread strategy for active control of surface properties is based on coating the substrates with a stimuli-responsive layer. Polymer coatings are natural candidates for stimuli-responsive applications, given the large variety of functional and structural possibilities developed during the past century.¹ Along these lines, different chemical (e.g., pH, salinity) or physical (e.g., thermal, mechanical, irradiation) external stimuli have been explored.² Most of these applications aim to control the macromolecular conformation, to, *in fine*, tune a particular surface property. In this direction, the use of an external electrical stimulus appears as an interesting option if charged monomers are included in the polymer chain, as electrical signals are relatively easy to control, and they are fully compatible with many current technologies (e.g., microfluidics).^{3–5} However, an applied field can interact with the polymer coating (and the entire system) through a number of different paths, which must be considered to achieve a rational control of the interface. First, there can be a direct effect of the field on the macromolecular conformation: charged moieties, permanent dipoles, and polarizable groups will respond to an external electrical field by migration of charges and reorientation of permanent or induced dipoles. In some cases, large conformational transitions can be triggered by these changes (e.g., helix–coil transition in polypeptides^{6,7}). An external field can also interact with other species, significantly altering values of local pH or ionic strength, which may trigger macromolecular conformation changes. This effect can be particularly significant for the case of weak polyelectrolytes

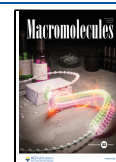
because of the dependence of the degree of ionization of the polyon on the chemical environment.^{8–11} In addition, there are a number of “chemical” changes that can be triggered by an externally applied electrical potential. Redox reactions can be driven by the applied potential, with charge and matter exchange at the level of the polymer coating, the electrode, or the environment.^{12,13} Some of these reactions may involve the solvent or low concentration species whose presence is difficult to avoid in concrete applications (e.g., dissolved gases). Furthermore, dissociation equilibria can be affected by a field strong enough, effectively shifting the ionization constant of a particular functional group (second Wein effect),^{14,15} which in the case of polyelectrolytes are often accompanied by important conformational changes. Other less desirable effects include electrical breakdown and irreversible rupture of chemical bonds of a particular coating by redox reactions. To further complicate matters, often a number of the above-mentioned mechanisms act simultaneously.

Electric field control of the conformation of surface grafted macromolecules has been explored in the past. A limited number of experimental studies have been reported.^{16–20} On the contrary, a larger number of theoretical^{3,4,21–23} and numerical^{24–27} studies have investigated the field-induced response of weak or strong polyelectrolytes (with variable or fix degree of

Received: November 17, 2021

Revised: February 11, 2022

Published: March 15, 2022



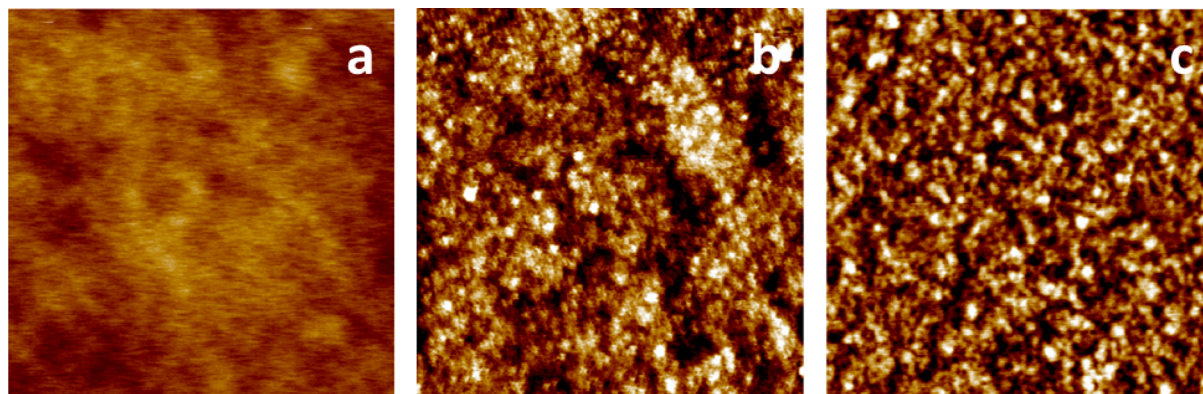


Figure 1. AFM height micrographs (500 nm $\times$ 500 nm) of (a) bare gold (tapping mode in air), (b) pAA-coated gold (peak-force mode in water), and (c) pDMAEMA-coated gold (tapping mode in air). A color scale range of 2.6 nm was used for the three images.

ionic association). Many of these studies involved the charge density at the electrode, σ_e , as the control parameter (noteworthy exceptions were reported in refs 3, 4, and 21). However, in practice, it is not possible to control this quantity. To tune the molecular conformation of grafted macromolecules, it would be desirable to control the potential difference between the supporting electrode and the bulk solution, ΔV_{we} . Nevertheless, control of this variable is also out of reach. It is natural to think about a single interface, but as is well-known in electrochemistry, the accessible quantity for an experimentalist is the potential difference between the electrode of interest (working electrode, WE) and a second electrode (reference electrode, RE), ΔV , ideally placed nearby. Thus, in most cases, it is not possible to know a priori the actual potential difference on a single electrode. ΔV combines the potential drop on both electrodes, and therefore the influence of the second electrode (e.g., chemical reactions and potential drop occurring through this interface) has to be integrated in the description of the system. This factor can be very consequential when small potential differences are at play. As long as the potential difference between the reference electrode and the bulk solution, ΔV_{ref} , remains invariable, any variation in ΔV directly involves as a change in ΔV_{we} . However, the choice of the RE is not without consequences. Its electric potential can evolve with time due to different chemical or physical processes. In addition, it is likely to change if any current pass through the system, for instance, as a consequence of a Faradaic process. Furthermore, when a nonzero current circulates, the ohmic iR potential drop will contribute to the measured ΔV . To avoid these technical complications, a three-electrode configuration is often implemented by using a potentiostat.¹² With this device the electrical potential of the WE is set with respect to the RE through a large impedance path, so that an extremely small current circulates and the potential of the RE remains reasonably constant. The circuit is completed by using a third (counter) electrode CE: the electrical current will flow through this electrode and the WE. Although it is often ignored, important changes in electric potential and chemical reactions happen at the counter electrode as a consequence of the passing current.

Another factor that must be considered is the nature of the WE. A metal in contact with an aqueous media will probably acquire a certain electric charge due to ionization, ionic adsorption, or even solvent polarization, which depends on the precise environmental variables (e.g., pH and ionic strength). In addition, depending on the applied potential, different electrochemical reactions may occur, with important

consequences for the operation of the system. The scenario can be more involved for functionalized electrodes, as the ones investigated in this work. As mentioned above, the ionization degree of weak polyelectrolytes depend on the environment; changes in the ionization degree of grafted polyion chains will modify ΔV_{we} . Attempts to change ΔV_{we} may induce changes in the ionization degree of the grafted chains, with a feedback effect on ΔV_{we} . It is clear that the interplay between the externally applied potential and the effective charge density on the electrode σ_e may complicate the control of the configuration of the polymer brush.

In this work we have investigated the electroresponsive character of brushes of weak (pH-sensitive) polyelectrolytes PE grafted on gold electrodes. For this purpose, we have used a three-electrode system in a quartz crystal microbalance to explore the influence of an externally applied electric field on the macromolecular coatings. We present a systematic exploration of the response and temporal evolution of surface-grafted polyelectrolytes to variations in the bias potentials of the supporting metallic electrode under different environmental conditions.

EXPERIMENTAL SECTION

Materials: Polymer Preparation. Disulfide-functionalized poly(acrylic acid), (PAA₄₇)₂S₂, was prepared by deprotection of the *tert*-butyl groups from poly(*tert*-butyl acrylate disulfide, (PtBA₄₇)₂S₂). The preparation of (PtBA₄₇)₂S₂ was performed as follows: 0.026 g (0.18 mmol) of Cu(I)Br was placed in a Schlenk tube. Next, 2.207 g (17.22 mmol) of deoxygenated *tert*-butyl acrylate (tBA), 0.041 g (0.24 mmol) of *N,N,N',N'',N'''*-pentamethyldiethylenetriamine (PMDETA), 0.054 g (0.12 mmol) of bis[2-(2'-bromoisobutyryloxy)ethyl] disulfide, (BiBOE)₂S₂, and 1.5 mL of deoxygenated acetone (5:3 by volume monomer/solvent) were added to the tube. All the reactants were added under nitrogen flow. The reaction was performed under a nitrogen atmosphere at 50 °C for 6 h. Then, the reaction mixture was diluted with chloroform and passed through an alumina column to eliminate the copper catalyst. The polymer was precipitated with methanol, dried, and characterized by GPC and ¹H NMR. To deprotect the (PtBA₄₇)₂S₂ and obtain the desired (PAA₄₇)₂S₂, 0.5 g of poly(*tert*-butyl acrylate) disulfide, (PtBA₄₇)₂S₂, was diluted in dichloromethane (CH₂Cl₂) in a round flask. Then, 2.23 g (19.52 mmol) of trifluoroacetic acid were added, and the mixture was stirred for 72 h at room temperature, forming a white powder precipitate. The solvent was removed by filtration, and the product was washed five times with dichloromethane to eliminate the acid residues. *M*_n: 12500 g/mol; *M*_w: 15100 g/mol (polydispersity, PD: 1.21).

Synthesis of Poly(2-(dimethylamino)ethyl methacrylate) Disulfide, (PDMAEMA₈₈)₂S₂. 3.732 g (23.74 mmol) of Cu(I)Br

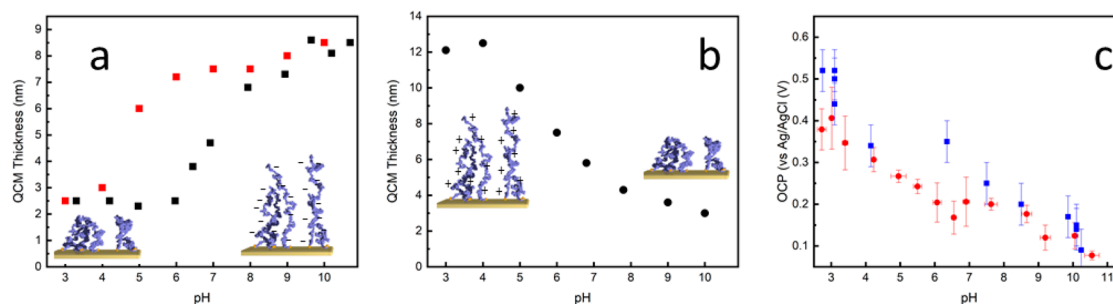


Figure 2. (a, b) pH dependence of the thickness of PE films calculated from changes in Δf and ΔD of the coated quartz crystal as a function of pH. (a) pAA. Black squares $[KCl] = 2$ mM. Red squares $[KCl] = 100$ mM. (b) pDMAEMA, $[KCl] = 2$ mM. (c) OCP for bare (blue squares) and pAA-coated (red circles) gold electrodes as a function of pH. pAA values are the average of the results obtained with 14 independently substrates. The gold values are the average of five independent substrates, measured after immersion in water for more than 2 h.

was placed in a Schlenk tube. Next were added 0.113 g (0.65 mmol) of N,N,N',N'',N''' -pentamethyldiethylenetriamine (PMDETA), 0.147 g (0.33 mmol) of bis [2-(2'-bromoisobutyryloxy)ethyl] disulfide, $(BiBOE)_2S_2$, and 2.5 mL of tetrahydrofuran (THF). Finally 3.732 g (23.74 mmol) of 2-(dimethylamino)ethyl methacrylate, DMAEMA (5:3 by volume monomer/solvent), was added. All the reactants were added under nitrogen flow. The reaction was performed under a nitrogen atmosphere at 60 °C for 2 h. Then, the mixture was diluted with chloroform and passed through an alumina column to eliminate the copper catalyst. The polymer was precipitated with hexane, dried, and characterized by GPC. M_n : 27800 g/mol; M_w : 33300 g/mol; PD: 1.20.

Polymer Brush Assembly. Disulfide-functionalized PEs were used to produce brush layers on gold surfaces by a “grafting to” approach. This method, based on spontaneous self-assembly triggered by readily formed gold–sulfur bonds,²⁸ has been broadly used before to coat metals with a large variety of compounds, as it is relatively easy to implement. It has been successfully applied before in studies of polymer brushes of pAA,^{29–31} pDMAEMA^{32,33} and others.^{34–37} Gold-coated quartz crystals (QXS 301, Q-Sense) were thoroughly cleaned and immersed in 1 mg/mL $(PAA_{47})_2S_2$ or $(PDMAEMA_{88})_2S_2$ solutions in ethanol for 24 h to enable the chemical grafting of the thiol-modified polymers on the surface and attain high coverages. After that period, the surfaces were rinsed with pure ethanol to remove nongrafted material and dried with dry nitrogen. To illustrate the extension of the grafting density achieved by self-assembly from ethanol, we have also prepared samples of both polyelectrolytes on atomically smooth gold surfaces prepared by mica templating, as described by Hegner and co-workers.³⁸ Atomic force microscope (AFM, Icon, Bruker) micrographs measured on these samples are presented in Figure 1. As can be observed in the height images, dense and homogeneous coatings of both polyelectrolytes are obtained in gold by self-assembly.

In a series of QCM experiments, octadecanethiol ($C_{18}SH$) was coadsorbed with the pAA polymer from a solution of 1 mg/mL $(PAA_{47})_2S_2$ and 0.02 mg/mL $C_{18}SH$ in ethanol (1:50 $C_{18}SH$:pAA in wt %) to minimize charge transfer across the metal–solution interface. For all samples, cleaning of the gold surfaces before functionalization involved steps of rinsing with Milli-Q water and ethanol and drying with nitrogen gas. Then the substrates were cleaned with an ultraviolet ozone cleaning system (UVOCS INC, model T0606E) for 30 min, rinsed again with ethanol, and dried with a filtered, dry N_2 flux. Polymer grafting was started right after the cleaning process was completed. Otherwise, a noticeable decrease in the quality of the functionalization (with less dense grafting) was observed. In all, we investigated 14 samples of pAA and 11 samples of pDMAMEA-coated surfaces.

Quartz Crystal Microbalance with Dissipation Monitoring (QCM-D). We investigated the conformational changes of grafted PEs with a quartz crystal microbalance with an electrochemistry module (QCM-D E1 and QEM 401, Q-Sense). The principles of this technique have been extensively described in the literature.³⁹ Briefly, the main shear-mode resonance frequency f of a quartz resonator and a number of odd harmonics are measured; a change in the effective mass of the resonator due to material adsorption/desorption translates into a

variation in the resonance frequency, Δf . In addition, the damping of the oscillation of the crystal is measured and used to calculate the “dissipation factor”, D , defined as the inverse of the quality factor of the resonance peaks.⁴⁰ A large value of D indicates quickly decaying oscillations of the crystal after the excitation is stopped, which is regularly observed for thicker and less rigid layers on the substrate. Thus, the measured Δf and ΔD can be related to the thickness and viscoelastic properties of the material adsorbed on the quartz crystal by using adequate models.³⁹ However, the physical parameters obtained from the fitting are often critically dependent on the nature of the model used and must be read with caution.⁴¹ Despite these liabilities, QCM-D is a fast, simple, and reliable technique to monitor the swelling degree of the polyelectrolyte brushes. Even though an accurate description of the polymer density on the substrate is out of reach by the information provided by QCM-D alone, this technique allows the qualitative exploration of the behavior of these systems under a large number of environmental conditions in a reasonable time. By comparison of the measured resonance frequencies of the crystal with the values measured at extreme pH values (when the polyelectrolyte brush is fully swollen or fully collapsed), an accurate assessment of the polymer degree of swelling is readily obtained.

We used 5.0 MHz quartz resonators covered with gold electrodes, which were coated with PE brushes as described above. At the beginning of each experiment, the coated resonator was placed in the measuring cell, and the resonance frequencies of the different odd harmonics were determined in air. Then, it was immersed in Milli-Q water adjusted to the pH and salt concentration to be studied for at least 30 min, until a stable baseline was established. This enabled the system to thermally equilibrate. For each experiment, we verified that the jump in the resonance frequency of the different harmonics upon immersion in water corresponded with the values predicted by Kanazawa and Gordon's equation.⁴² Otherwise, the sample was discarded. The effect of pH, ionic strength, and applied potential of the gold surface on the polymer conformation was investigated by measuring the response of the odd harmonics ($n = 3$ to $n = 13$). The pH-responsiveness of the prepared samples (the jump in Δf of the different harmonics upon change from pH 3 to pH 10) was used as a control criterion for the quality of the different samples investigated. In all experiments conducted, pH was varied by adding low-volume aliquots of concentrated KOH or HCl solutions. We avoided using buffer solutions to control pH, as they regularly contain multivalent ions which may introduce ion-specific effects, complicating the interpretation of the experimental results. Electrical potentials were applied to the Au substrates vs an Ag/AgCl reference electrode (DRIFEF-2SH, World Precision Instruments) by using a CHI650D electrochemical workstation (CH Instruments). All measured potentials herein are reported vs an Ag/AgCl RE.

RESULTS

Weak Polyelectrolyte Brushes: pH Responsiveness.

The influence of environmental variables on the properties of PE brushes has been extensively investigated in the past.^{8–11} For the

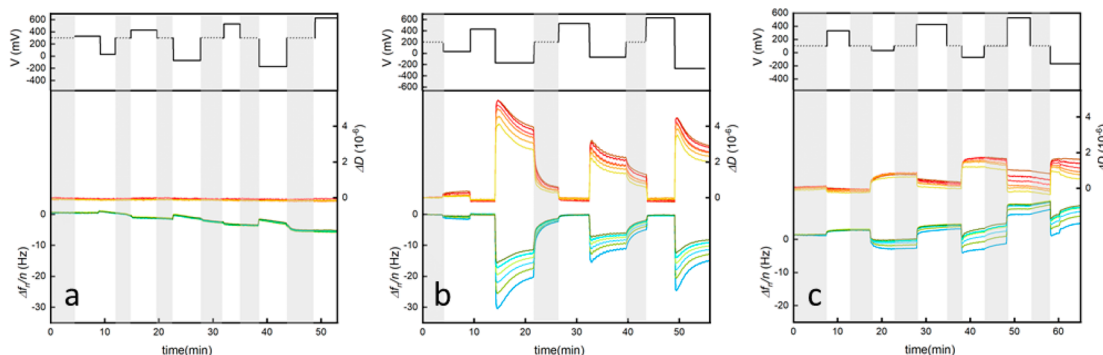


Figure 3. Δf and ΔD for the different odd harmonics for pAA-coated gold electrodes for different applied potentials between working and reference electrodes (upper panels). [KCl] = 2 mM; (a) pH 3.0, (b) pH 4.5, and (c) pH 10.0. The gray areas correspond to time lapses when the potentiostat was inactive.

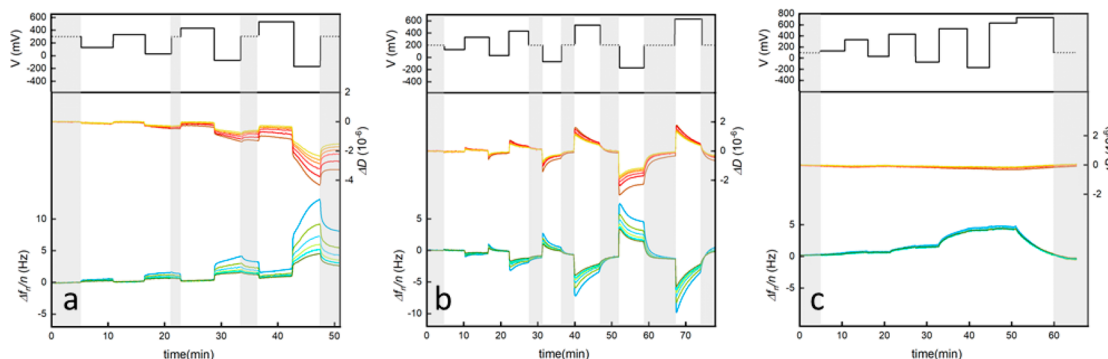


Figure 4. Δf and ΔD for the different odd harmonics for pDMAEMA-coated gold electrodes for different applied potentials between working and reference electrodes (upper panels). [KCl] = 2 mM; (a) pH 4.0, (b) pH 7.0, and (c) pH 10.0. The gray areas correspond to time lapses when the potentiostat was inactive.

samples studied in this work, we regularly explored the changes on the conformation of the grafted brushes induced by changes in pH, before evaluating the influence of an applied external potential. Increasing pH triggers the ionization of the carboxylic groups in polyacid (e.g., pAA) chains or the deprotonation of the tertiary amino groups in polybases (like pDMAEMA), with distinctive effects on their conformation. Examples of the pH-induced variation of effective brush thickness, T (measured by QCM-D), are presented in Figure 1. To obtain these results, the variations in Δf and ΔD of the different odd harmonics measured by QCM-D ($n = 3$ to $n = 13$) were used to calculate T by using the QTM software developed by Johannsmann,⁴³ assuming a mass density of 1 g/cm³. As expected, the extension of the pAA brush augments with increasing basicity due to the increasing intrachain electrostatic repulsion triggered by enhanced ionization (Figure 2a). The opposite is true for the pDMAEMA chains (Figure 2b). In general, relatively fast changes were observed; typically, several minutes after the change of pH, a stationary film thickness state was achieved. We can define an apparent pK_a for the PE studied, as the inflection point of the measured T vs pH curves. For the case of pAA (Figure 1a), our results show a clear dependence of pK_a with salt concentration: while at high concentration the effective pK_a is similar to the value observed in bulk solutions (pH ca. 4.5), the transition is acid-shifted around 2 pH units at low ionic strengths. This shift in pK_a is due to the unfavorable effect of the long-range electrostatic interaction on monomer ionization, which is more important at low ionic strengths due to the increasing electrostatic screening length. An acid shift in pK_a of grafted pAA brushes has been reported previously by other

groups;^{44–47} we have extensively discussed parallel results obtained with pAA brushes prepared by a different method in a recent publication.⁴⁸

Polyelectrolyte Brushes: Electroresponsiveness. It is not an easy task to locally regulate the values of pH or salt concentration to control the conformation of grafted polyions. Actuation on these variables often involves mass transport, requiring mixing or diffusion control, and entails relatively long response times and global scales. On the contrary, an applied electric field may be a practical method to trigger local PE response, as the field is relatively easy to control and implement, and these molecules contain ionized monomers promptly affected by it. Before exploring the electroresponsive character of the PE brushes, it is necessary to determine the equilibrium value of the potential difference ΔV between the coated WE and the RE in the absence of any external perturbation (open circuit potential, OCP). Determining the OCP is necessary to have a measure of the magnitude (and the direction) of the perturbation applied to the grafted layer when a different potential value is imposed on the metallic electrode: it is generally wrong to assume an (unperturbed) equipotential state between both electrodes.

The influence of pH on the OCP values measured for a pAA-coated electrode is shown in Figure 1c. The OCP is determined by the characteristics of the metallic electrode (which usually evolves with time) and the charges introduced by the grafted polyions. As can be observed in the figure, the OCP of bare gold is pH-sensitive, a feature that is observed for the vast majority of aqueous interfaces. Large potential variations are observed when pH is far from neutrality due to preferential adsorption on the

electrode of the more concentrated ion (H_3O^+ in acid and OH^- in basic conditions). The pAA coating induces a negative shift in OCP (smaller than 100 mV) with respect to the bare electrode.

In the first set of experiments described (Figures 3 and 4), we applied a number of discrete steps in the potential between the gold electrode and the RE, alternated with disconnection stages (open circuit condition). The potential steps imposed drive the system out-of-equilibrium; thus, the conformation of the grafted polyelectrolytes relaxes to reach the new equilibrium state under the applied field. The evolution of the systems is explored by recording the response in the signals of the QCM-D and the current circulating through the circuit. To avoid electrochemical reactions of the polymer coating (i.e., reductive degradation of the Au–S bonds), we restricted the range of applied potentials from -0.3 to 0.5 V, a “harmless” electrochemical range. Typical examples of the measured traces are reproduced in Figures 3 and 4. To simplify the visualization of the results, in these figures we have set the values of Δf and ΔD as zero in the condition before any external potential is applied (“open circuit” condition; time equal zero on each trace). The open circuit potential (potential difference between the working and reference electrodes, OCP) at the particular conditions of pH and ionic strength (measured before application the potential steps) is indicated by dotted lines. The periods when the action of the potentiostat were discontinued (lapse of time between two imposed potential steps, when the circuitry of the potentiostat is bypassed) are highlighted as gray areas. The evolution of the QCMD signals can be qualitatively understood as follows. A reduction in the resonance frequency of the coated crystal ($\Delta f < 0$) indicates an augmentation of its effective mass. In parallel, increasing the width of the resonance peak ($\Delta D > 0$) implies a more significant energy dissipation in the damped oscillation. Both processes are identified with increasing swelling of the polymer brush. On the contrary, increasing Δf and decreasing ΔD may be expected for shrinking polymer brushes. The potential-induced response of the systems depends on the physicochemical environment (pH and ionic strength) which determines the initial conformation of the polymer brush. In general, three distinctive regimes were identified:

- (i) Under conditions of small charging density (low pH for pAA, Figure 3a, or high pH for pDMAEMA, Figure 4c), only very minor changes in Δf (< 5 Hz) and ΔD ($< 1 \times 10^{-6}$) of the functionalized resonators were observed for the different applied potentials, within the range explored. These small variations cannot be differentiated from typical potential-induced shifts observed with bare gold-coated quartz crystals.⁴¹ A characteristic example of the response of an uncoated crystal is reported in the Supporting Information (Figure S1).
- (ii) Under conditions of significant ionization (low pH for the case of pDMAEMA, Figure 4a, or basic conditions for pAA, Figure 3c), electrostatic interactions triggers enhanced extension or collapse of the polymer brushes. Thus, increasing the potential between the WE and the RE prompts the increase (for the case of pDMAEMA) or the decrease (for pAA) of the effective thickness of the polymer brush. The opposite scenario is observed upon lowering the applied potential of the gold electrode, which triggers a repulsive (attractive) force between the electrode and the polyanion (polycation). As can be observed in Figures 3 and 4, the potential-induced changes in polymer brush state are neither immediately

nor completely reversed under removing the action of the potentiostat (with the electrode slowly returning to OCP). Significant irreversibility and hysteresis are present in the response to potential steps. It is often necessary to impose a significant potential reversal (a potential value larger or smaller than the originally measured OCP) to trigger the recurrence of the polymer brush to its previous condition. This aspect, which must be considered for potential applications of these polyions as smart coatings, will be further discussed below.

- (iii) A more complex scenario is observed under conditions close to pK_a of the grafted polyions (Figures 3b and 4b). In this case, a nonmonotonic, slowly evolving response is generally observed when the electrode potential is changed, implying significant changes in the configuration of the polymer chains, as can be observed in Figures 3b and 4b. For instance, applying a negative (positive) potential step to a pAA (pDMAEMA) brush when the polyions are partially (or completely) uncharged initially triggers a fast swelling followed by a slow partial shrinking of the polymer chain. The opposite sequence was observed when a positive (negative) potential step is applied to partially charged pAA (pDMAEMA) brush. In that case, the potential change triggers initial shrinking, followed by a partial recovery of the effective thickness of the polymer brush. Typically, after an initial fast response (first few seconds after potential change), the evolution of the PE layer is much slower, taking few minutes to achieve what appears to be the stationary state. A two-stage (fast–slow) nonmonotonic response of pAA brushes to applied potential steps, similar to what is observed in this work, was reported by Borisova and co-workers.¹⁶ The initial potential-induced swelling of the polymer brushes can be very significant: depending upon the initial conditions and the applied potential step, it can add up to the changes observed throughout the complete PE ionization. On the contrary, the net change after the final, steady-state condition is reached is less important; in some cases, the PE brush even returns to the same state as before the application of the potential step. In all cases, the change in polymer conformation is determined by the difference between pH and pK_a at the conditions of ionic strength investigated. Thus, for pAA at pH 4.5 and 2 mM KCl (more acidic than pK_a , Figure 3b), the swelling–shrinking sequence described above can be observed at sufficiently large negative potential steps. However, only minor changes are observed at positive potential steps (with respect to the OCP). On the contrary, for pDMAEMA at pH 7 and 2 mM KCl (very close to pK_b , Figure 4b) large (nonmonotonic) changes in polymer effective thickness are observed both for positive or negative potential steps.

In the experiments just described, we alternated stages in which the potentiostat was used to impose a potential value to the supporting electrode (with respect to the RE), followed by discharging periods. When a potential step is imposed, the system is “forced” to a state of higher energy compared to the open circuit state. Once the effect of the potentiostat is stopped, the potential difference will return to the original OCP by ohmic leakage between the electrodes. Some examples of the slow temporal evolution of the measured WE potential after imposing different potential steps are reported in the Supporting Information (Figure S2). After potential relaxation, it can be

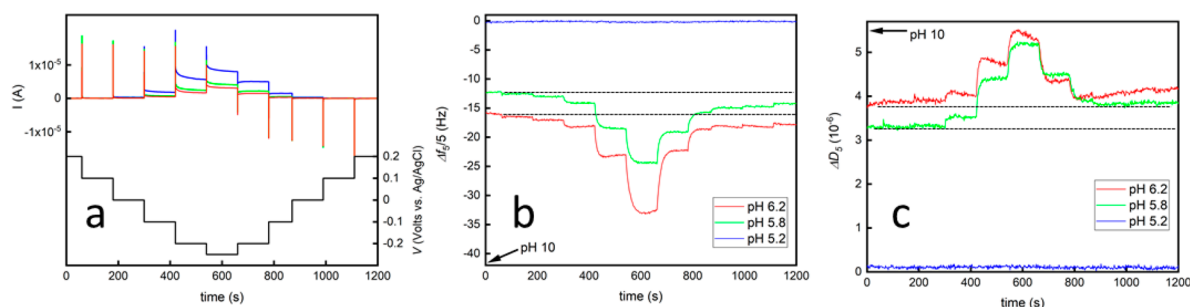


Figure 5. Response of a pAA-coated crystal to the application of a sequence of potential steps at different pH conditions. [KCl] = 2 mM. (a) Applied potential steps and measured current. Longer Faradaic currents are measured at more acidic pH. QCM-D signal of (b) Δf and (c) ΔD for the fifth harmonic. Similar traces were obtained for all odd harmonics. The arrow indicated the values of Δf and ΔD measured at pH 10 (fully hydrated pAA chains).

expected than the system should return to the low-energy (discharged) state. Nevertheless, it is important to distinguish between the global evolution of the potential difference between the WE and RE and the relaxation in the conformation of the grafted polymer chains, monitored by the QCM response. In our experiments, a complete reversal of polymer conformation upon discharging was not always observed. The conformational evolution of the systems during discharging was different for the two polymers investigated, depending also on the environmental conditions. When the PE was fully ionized before applying the potential step, partial (pDMAEMA; Figure 4a) or no recovery (pAA; Figure 3c) of the initial swelling condition was observed after stopping the action of the potentiostat. As mentioned above, a significant reversal in the applied potential (values beyond the OCP) was often necessary to fully revert the change triggered by the action of the initial potential step. On the contrary, in conditions closer to the ionization pH, the effect of the applied potential step was partial (pAA; Figure 3b) or completely reversed (pDMAEMA; Figure 4b) a few minutes after the potentiostat was turned off. Thus, when the applied potential entails important variations in the system conformation (i.e., pH close to pK_a), or for larger differences between the applied potential and the equilibrium OCP, a clear tendency to return to the unperturbed state is observed when the potentiostat is stopped. On the contrary, the driving force to return to the equilibrium state is weaker for small potential changes, or when the potential-induced response of the system is smaller (pH far from pK_a), and a more important hysteresis in the system response is observed. We will come back to these important issues at the end of the discussion.

Effect of Applied Potential: Faradaic and Non-Faradaic Processes. To investigate the reversibility of the potential-induced changes, it is probably more useful to impose an uninterrupted control on the potential of the metallic electrode during a charging–discharging cycle. Each time a potential step is applied, the system (including the grafted polymer brush) is driven to a new stationary condition, far from the original equilibrium state (in absence of the applied potential). As an example, the response of pAA-coated electrode to a sequence of small potential steps is presented in Figure 5, for different pHs of the surrounding aqueous bath. For the sake of clarity, we show only the results corresponding to the fifth harmonic, although similar results were obtained for the different odd harmonic measured. In this and subsequent figures the values of Δf and ΔD are referred to the data measured at pH 3, when the PE is expected to be mostly uncharged, and the effective thickness of the grafted brush will be the thinnest

achievable. Once again, the drop in resonance frequency f and the growth of D can be identified with increasing swelling of the polymer brush.

The sequence of steps in the applied potential between the WE and the RE and the passing current measured between the WE and the CE are shown in Figure 5a. As can be observed in the figure (panels b and c), for the applied potentials (between +0.2 and −0.2 V) significant changes in the PE conformation were observed at pH above 5.5. Application of increasingly larger negative potentials triggered PE swelling. On the contrary, no change was detected at pH 5.2, when the PE is mostly nonionized. Interestingly, when swelling is observed, at sufficiently negative potentials the effective thickness of the PE achieves values similar to the ones observed for fully charged PE layers (indicated by black arrows in the figure). Similarly to what was observed for larger potential steps, significant irreversibility is often found in a cycle of applied potential (see dashed lines in Figure 5b,c).

It is informative to analyze the current passing through the WE (and the CE) when the potential steps are imposed (Figure 5a). When the metal electrode potential is changed, electrical current flows through the electrode. In the absence of charge transfer across the metal–solution interface, an imposed voltage step implies a change in the charge density in the metallic electrode, which is accompanied by the accumulation of an identical charge of opposite sign in the solution side, nearby the electrode. In this scenario, in the absence of any electrochemical reaction (often designed as a perfectly polarizable electrode and a “non-Faradaic” process), one can consider the metal–solution interface as a capacitor, with a final charge density determined by the applied potential, which is reached after certain charging time. This double-layer charging/discharging process is commonly very fast. It corresponds to the short spikes in current observed after each step in applied potential. After this charging time, current ceases to flow through the circuit. For the data presented in Figure 5, this is the case for potential values between −0.1 and 0.2 V. On the contrary, for potentials more negative than −0.1 V, a nonzero current is observed after the initial current spikes, which typically remains non-null as long as the potential on the electrode is maintained. The presence of this finite current is an explicit signal that an electrochemical (Faradaic) process is occurring.

We can examine separately the initial response to the applied potential and the evolution toward stationary state conditions.

i. Short Time Response: Double-Layer Charging. As mentioned above, the double-layer response to an applied potential step is commonly a fast process. Typically, the non-

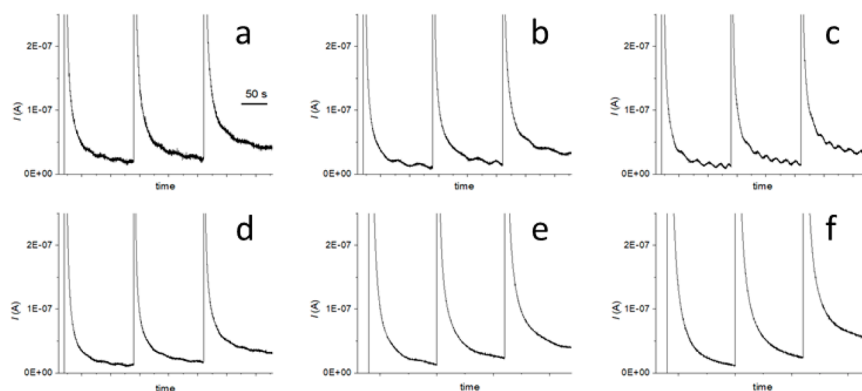
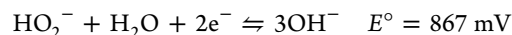
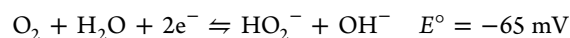


Figure 6. Current intensity upon application of a series of potential steps, for different pH conditions. The three potential steps sequence displayed on each panel correspond to (i) OCP + 0.1 V, (ii) OCP, (iii) OCP − 0.1 V, between the WE and the RE, related to the OCP at each pH condition. pH/OCP: (a) 3.4/0.3 V; (b) 5.3/0.25 V; (c) 5.6/0.24 V; (d) 6.4/0.21 V; (e) 7.5/0.19 V; (f) 10.1/0.08 V. The current axis is limited to 0.25 μA to make visible the long time oscillations observed (I reached values up to ca. 25 μA after each potential step).

Faradaic charging current decays exponentially with time, with a decay RC time determined by the solution resistance and the double-layer capacitance. However, the process is more subtle in the presence of the PE layer, in particular if the charge density of the PE is not constant (potential dependent). By using a potentiostat, it is the potential difference between the WE and the RE which is measured and controlled. However, as can be anticipated, the extension of the charging of the PE contributes to this potential difference. This contribution (of the order of 0.1 V) can be significant, in particular if relatively small potential steps are considered. An instructive discussion of this aspect was recently reported by Borisov and co-workers.⁴ Changes in the ionization degree of the PE during the charging process will certainly affect the evolution toward the desired potential. Some examples of the measured current during the short-time process of charging (upon several applied potential steps) are presented in Figure 6. At pH conditions far from pK_a ($\text{pH} < 4.5$ or $\text{pH} > 6$), the current measured upon a particular potential steps decays monotonically with time. On the contrary, a distinctive oscillatory pattern can be identified when the pH of the surrounding solution is close to pK_a . This nonmonotonic response in the final stage of the charging process is likely related to charging–discharging cycles of the PE layer, which affect the potential of the WE as described above and therefore the flow of charges to/from the metallic electrode and through the potentiostat.

ii. Long-Time Response: Faradaic and Non-Faradaic Processes. At sufficiently large negative potentials (< -0.1 V) a non-null current was observed after the initial charging spike (Figure 5a). In these cases, Faradaic and non-Faradaic process occurs simultaneously. The Faradaic current is determined by the diffusion of the species involved in the process and the speed of the chemical reaction. As can be observed in Figure 5, the observed current depends on the pH of the solution, being more important at more acid conditions. To achieve a complete understanding of the behavior of the system, it is necessary to identify the electrochemical reactions occurring and its possible consequences on the behavior of the PE. The two polyelectrolytes investigated are not electrochemically reactive within the range of potentials investigated in this work (-0.3 to 0.5 V vs Ag/AgCl). Water is also electrochemically stable in this regime. On the contrary, cathodic reduction of dissolved oxygen can be triggered at sufficiently low potentials. Although the reaction mechanism of this reaction is still not completely elucidated, it is

generally accepted that it involves 2 or 4 e^- pathways (reduction to peroxide or direct reduction to OH^-), proceeding in basic media as^{49,50}



The actual reaction mechanism depends on pH and on the details of the substrate. The 2 e^- pathway (the first reaction above) appears to be the more relevant for polycrystalline gold electrodes.⁵¹ In any event, this reaction results in generation of hydroxide ions (or hydronium ions consumption), significantly altering the local pH at the level of the gold electrode. Thus, when a sufficiently low applied potential is imposed on the gold electrode, the basicity of the nearby solution is enhanced, promoting the swelling (deswelling) of grafted polyacid (polybase) brushes. Large shifts in local pH can be generated by this reaction, which can trigger important changes in the conformation of grafted weak polyions. A number of reports illustrating the possibility of using this stimulus for controlled drug release,^{52,53} blocking of electrodes,⁵⁴ or local control of substrate mechanical properties⁵⁵ have been recently reported.

Hindering Oxygen Reduction. To evaluate the impact of the contribution of the reaction of oxygen reduction, we implemented two strategies to limit its advancement. First, we deoxygenated the solution in contact with the PE brushes by extensive Ar gas bubbling and evaluated the response of the grafted polymer brush to potential steps under an atmosphere rich in Ar. A second strategy involved the coadsorption of an alkanethiol on the gold electrode during polymer grafting to block the access of the oxygen to the electrode, limiting charge transfer and the progression of the reaction.

The response of a pAA-coated gold electrode to a sequence of potential steps between 0.15 and -0.25 V in air or under Ar atmosphere is presented in Figure 7. At the chosen solution pH ($\text{pH} = 6$), the pAA brush was close to uncharged/contracted conformation ($\Delta f = 0$ Hz) in both cases. However, as increasingly reductive potentials are imposed on the gold electrode, swelling of the polymer brush was readily detected. As could be anticipated, the magnitude of the cathodic current and the swelling of the polymer brush were more important when the experiment was conducted under regular air atmosphere. However, significant brush swelling is also observed under Ar environment. It is also noteworthy that while at low applied potentials the system appears to reach a stationary state in both

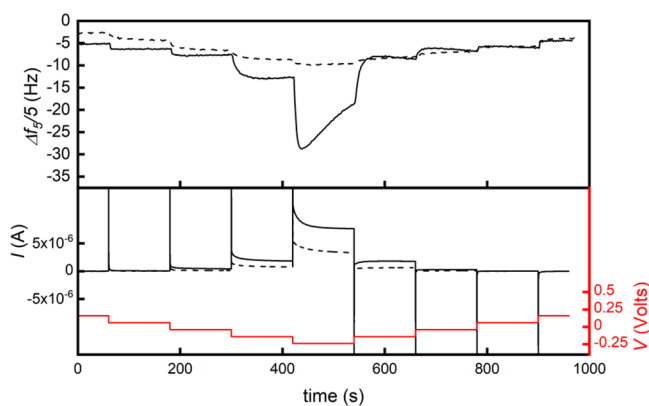


Figure 7. Response of a pAA-coated WE under air (full lines) and Ar (dashed lines) atmosphere. [KCl] = 2 mM, pH 6.0. Upper panel: changes in Δf for the fifth harmonic. Similar traces were obtained for all odd-harmonics. Lower panel: sequence of potential steps and measured current.

cases, at larger negative potentials the thickness of the polymer brush reaches a maximum value and then shrinks only for the system under air (as described above). This point is further discussed below. Analogous results were obtained when oxygen reduction was blocked by partial functionalization of the gold electrode with $C_{18}SH$ (Figure S3).

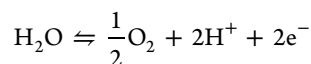
The results just described evidence the significant influence of oxygen reduction and the subsequent change on local pH on pAA conformation. It is apparent that this reaction provides an interesting path to fine control of the conformation of the polymer brush, as has been suggested in previous reports.^{52,53,55} However, it is also apparent that this reaction is not the only factor controlling the electroconformational changes observed; even when it is blocked, significant potential-induced changes in the PE chains are observed. These rearrangements are a consequence of the motion of ions and polymer chains, driven by the applied field.

For possible applications of this control path, it is also important to explore the long-time behavior of the PEs. Figure 8 shows the response of a pAA brush to large positive (0.2 to 0.6 V) and negative (0.6 to -0.2 V) potential steps. After the positive step, a significant reduction in effective thickness of the polymer layer is observed, followed by a minor change in the opposite direction. Few tens of seconds after the application of the positive potential step, the system appears to reach a stationary state, which is closer to the condition observed at

acidic pH. The response is different upon the negative potential step. Initially, a fast and large variation of the effective thickness is observed, with the polymer layer reaching a state similar to the one observed under basic pH conditions. However, a slow shrinking of the brush follows this swelling. The stationary condition, similar to the unperturbed state, is achieved after few hundred seconds. The slow evolution of the polymer conformation suggests that material diffusion controls the kinetics of the process. Assuming that charge transfer kinetics at the electrode is very rapid, the reaction progression will be limited by the diffusion of species involved. A crude estimation of the concentration profile of oxygen and hydroxide in the experimental cell can be performed by solving Fick's diffusion equation, as has been extensively described in the literature.¹²

Some estimates of the progression of this profile are presented in Figure 8b, considering that initially the concentration of oxygen was 0.25 mM (standard concentration of oxygen in water in contact with air) and the global pH was 7. As can be observed in the figure, the OH^- concentration close to the electrode increases significantly soon after the electrochemical reaction starts, reaching concentrations corresponding to pH ca. 10; this basic pH condition will trigger the ionization of the grafted pAA. However, after few tens of seconds, the diffusion layer (region near the electrode with an oxygen concentration significantly lower than bulk concentration) reaches ca. 1 mm, of the order of the separation between our working and counter electrodes (0.8 mm). After the electrochemical reaction significantly depletes the oxygen initially present in the reaction cell, further progression of the electrochemical reaction will be limited. Under this scenario, pH will shift toward more acidic values after diffusion of OH^- from the region near the electrode.

The precise features of the temporal evolution of the process just described are necessarily determined by the behavior of the global system. One aspect often neglected is the influence of the reaction that takes place at the counter electrode (CE), which "closes" the circuit in the electrochemical cell. In our particular case, it is likely that when oxygen is reduced at the WE, water oxidation will occur at the surface of the CE



Hydronium, which is produced during this reaction, will also diffuse, further neutralizing the pH change induced by the reduction of oxygen described above. Ultimately, the details of the pH evolution in the cell will depend on the precise characteristics of the cell: size, shape, and nature of the different

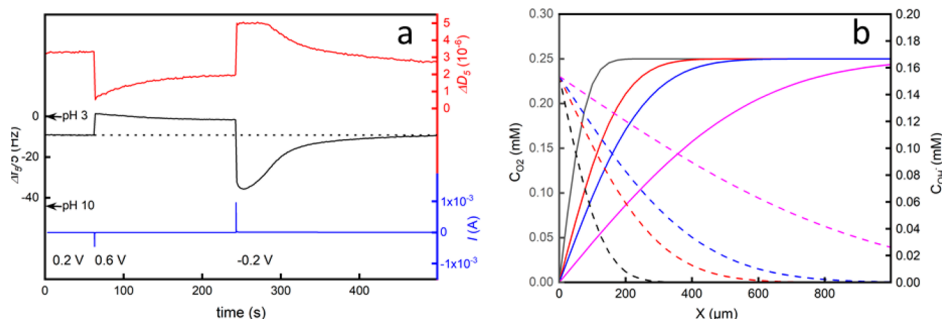


Figure 8. (a) Response of a pAA-coated electrode to steps in the applied potential. [KCl] = 2 mM, pH 6.8. Upper curves: changes in Δf and ΔD for the fifth harmonic. Lower curve: measured current. The Δf values at pH 3 and 10 are indicated by the arrows. (b) Calculated concentration profile of dissolved oxygen (full lines) and hydroxide ion (dashed lines) after application of -0.2 V between CE and RE for 1 s (black), 5 s (red), 10 s (blue) and 50 s (magenta), considering that the solubility and diffusion coefficient of oxygen in water are 0.25 mM and $2.01 \times 10^{-9} \text{ m}^2/\text{s}$.

electrodes. In particular, the limitation of the amount of oxygen present to drive the electroconformational changes in the electrode must be considered for any potential application of this control path, especially if small volumes are involved (e.g., in microfluidic devices).

History Effects: Memory and Partial Irreversibility.

Another important aspect to engineer responsive PE coatings is related to the reversibility of the response of the system. As can be observed in the Figure 5, the values of Δf and ΔD after a cycle of applied potentials are not exactly the same as before the cycle. Similar lack of full reversibility can be observed in some of the results presented in Figures 3 and 4. Generally, we found that the farther the system was driven away from the equilibrium condition (in absence of the external potential), the larger was the irreversible change after removal of the perturbation. This effect was more important when substantial swelling was induced by the applied potential, but we observed it both during swelling and shrinking. To illustrate this point, the response of a grafted pAA brush to potential steps of different magnitudes are presented in Figure 9. For the case of pH 6.1

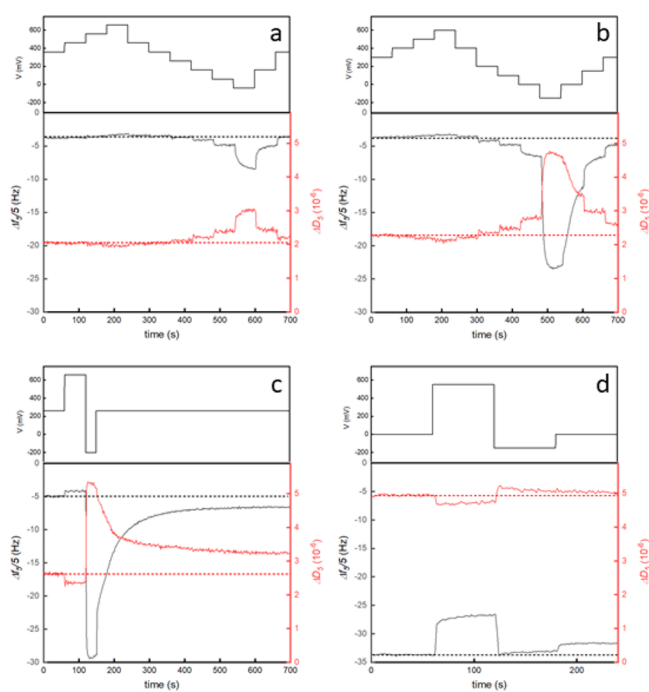


Figure 9. Response of a pAA-coated electrode for different sequences of applied potential steps. [KCl] = 2 mM; (a–c) pH 6.1; (d) pH 10.0. Upper panels: sequence of applied potential steps. Lower panels: changes in Δf and ΔD for the fifth harmonic. The minimum applied potential was (a) -40 , (b) -150 , and (c) -200 mV.

(Figure 9a–c), applying increasingly larger negative potentials results in increasingly larger swelling and incomplete reversal in polymer conformation after returning to the OCP condition (the value of potential at the beginning of each cycle). Similar but less significant effects were observed upon applying positive potentials with respect to the OCP value at basic pH (Figure 9d). In the example displayed, a potential value more negative than the initial OCP needs to be applied to reverse the shrinking effect of the large positive steps (130 s); on the contrary, after returning to the initial potential value, the system acquires a less swollen configuration (180 s).

Effect of Ionic Strength. All the conformational changes described in this work involve electrostatic interactions. It is reasonable to anticipate that the ionic strength of the media will affect the behavior of the system because of its influence on the screening of the electrostatic interaction. For instance, the ionic strength has an important impact on the PE swelling and the value of pK_a , as described before. We explored the response of a pAA-coated electrode at two different salt concentrations. A cyclic voltammetry CV strategy was implemented to perform a faster screening of the effect of the applied potential. In a CV experiment the potential between the WE and RE is linearly ramped between two limit values via a triangular potential variation, while the current flowing between the CE and the WE is measured. Typical results of these experiments are presented in Figure 10. As can be observed in the figure, changing the

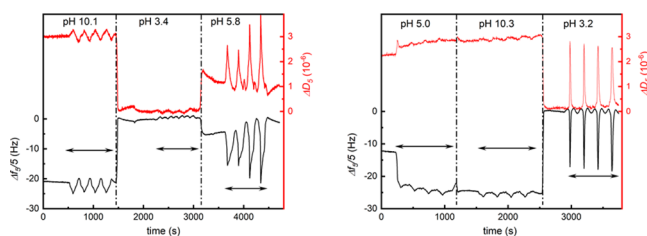


Figure 10. CV results. Temporal evolution of Δf and ΔD for the fifth harmonics for pAA-coated gold electrodes at different pH (as indicated). A triangular potential wave between -0.25 and 0.4 V was applied between the RE and the WE. Scan rate 0.01 V/s. Arrows indicate the period of time when the potential was applied. [KCl] = 2 mM (left) and 100 mM (right).

concentration of salt alters significantly the electroconformational changes of the polymer brush. At basic pH values, a cyclic behavior in polymer swelling, which follows the potential ramp, can be observed. However, increasing I induces a substantial reduction of the polymer response for a given potential excitation. This is a consequence of the enhanced electrostatic screening at larger values of I : the motion of the ions in solution counteracts the effect of the applied electric field, reducing its effect on the polyion. The higher the concentration of salt, the smaller will be the field-induced conformational changes of the polyion. For the example presented in Figure 10, the observed change in effective thickness of the polymer layer was 3 times larger at 2 mM than at 100 mM (0.9 nm vs 0.3 nm, assuming the validity of Sauerbrey's equation³⁹). On the contrary, at acid pH values we observed significant swelling–shrinking cycles of the pAA brush at large values of I , while no significant changes were observed at low salinity. The enhanced response at large ionic strengths is probably related to the acid shift in pK_a described before (Figure 2). It seems that the transient pH change triggered by oxygen reduction is enough to trigger the ionization of the polyion for the case of large I but not at low salt concentrations. Assuming again the validity of Sauerbrey's equation,³⁹ the observed Δf change (at large I) corresponds to a variation up to 4 nm in the effective polymer thickness, which is comparable to the maximum pH-driven changes obtained with the same polymer (cf. Figure 2). Finally, at intermediate pH, while we observed subsequent swelling–shrinking cycles for the case of low I , an irreversible voltage-induced swelling of the grafted brush was observed at high I after the first cycle, suggesting again the need of overcompensating the applied potential step to achieve a reversible response. In this regime, the behavior of the system will depend on the precise pH value, in

relation with pK_a . Nevertheless, in all cases complete swelling (shrinking) of the grafted polymer will be prompted if a sufficiently large negative (positive) potential is applied.

The different results presented in Figure 10 are a good summary of the diverse scenarios observed in this work. First, the conformation of fully ionized polyions can be manipulated by the applied field, mediated by the free ions, as can be expected from electrostatic considerations. Second, pH changes driven by oxygen reduction can trigger important ionization and conformational changes of the weak polyions. Ionization can also be directly triggered by the applied voltage. These changes are partially irreversible, requiring an excess potential change (in the direction opposite to the one that triggered the change) to return to the original state.

DISCUSSION

Conformational Changes: Electrochemistry vs Physical Stimuli (Field-Induced Dissociation and Counterion Motion). From the body of results described, it seems clear that electrochemically triggered changes in local pH can be used to control the conformation of pH-responsive polyions. However, this “chemical” driving force does not explain the whole set of observations reported in this work. For instance, it does not explain the behavior observed with pDMAEMA at pH close to pK_b (Figure 4b) or the potential-induced changes in pAA in the absence of Faradaic processes (Figures 3 and 7, Figure S3). In particular, it is intriguing that by applying a negative (positive) potential to a polyacid (polybase) brush, a significant increase in film thickness is promptly observed. Even though this potential-induced swelling was partially reversed in a later stage, the final stationary state of the brush was often more hydrated than the condition before the application of the potential step. In fact, it seems reasonable to expect the opposite tendency: more positive potentials on the metallic electrode should promote pAA ionization and pDMAEMA deprotonation; on the contrary, more negative potentials should lead to protonation of the polybase and stabilization of the carboxylic form of the polyacid. A number of studies compatible with this second scenario (ionization promoted by opposite charges between electrode and coating), theoretically described by Smith and White,⁶¹ have been reported.^{56–60} However, the opposite trend (the one observed by us) has also been found in different studies of electroresponsive weak PE brushes^{16,17} and in other studies of grafted acid/base groups.^{62–65} We do not have a conclusive explanation for this behavior. The possibility of field-induced dissociation of the weak polyions (in the line of the second Wien effect for weak electrolytes^{14,66,15,67}) cannot be discarded, as large electric field are imposed when a sudden change in potential is applied. Subsequent migration of counterions in response to the applied field could then explain the relaxation of the initial potential-induced swelling. Alternatively, a scenario controlled by the motion of the counterion cloud can be imagined. For instance, in the case of the polyacid, the application of a negative potential step induces a repulsive (attractive) interaction between the substrate and the polyion (counterions), favoring the swelling of the brush driven by the enhanced electrostatic repulsion between charged monomers. This step would then be followed by a slower deswelling, controlled by ionic diffusion from the external reservoir to the brush. Thus, the two-step response might be related to the different time scales of the response of ions in solution and grafted polyions, as argued by Borisova and co-workers.¹⁶

History Effects: Hysteresis and Irreversibility. Another aspect that deserves further consideration is the partial irreversibility and hysteresis observed in the electroconformational changes. Three different situations associated with this irreversibility were identified. First, the slow evolution of the OCP toward its equilibrium value (Figures 3 and 4) after the action of the potentiostat is discontinued. This global effect must be taken into account and may be of use in practical applications (e.g., for temporary gating). Second, a nonmonotonic response was often observed when a potential step was imposed (cf. Figures 3, 4, and 7, Figure S3). In some cases, this is due to the limited extent of the oxygen reduction that drives pH changes, as discussed above. However, this effect can not be argued in all circumstances; it cannot explain our observations with pDMAEMA or for positive potential steps for the case of pAA, as discussed in the previous paragraph. Finally, we regularly observed partial irreversibility in the behavior of the polymer brushes, as they failed to return to their original conformation after application of a cycle of potential steps. The application of an overpotential is often necessary for the system to return to its original state. This effect is more significant for larger excursions from the equilibrium states, in particular when the swelling of the brush is significantly enhanced by the application of the potential steps (Figure 9).

Several recent reports have discussed hysteresis and memory effects in the response of grafted weak polyelectrolytes upon pH changes. Using infrared ellipsometry spectroscopy, Aulich and co-workers observed significant hysteresis in ionization of grafted pAA in a pH cycle, with important consequence on the wettability of the system.⁴⁴ An analogous hysteretic response was reported by Weidman and co-workers in a study of permeability of pAA-coated membranes. Interestingly, they established that this hysteresis was related to the ability of pAA chains to hydrogen bond in the nonionized state.⁴⁶ In an extensive wettability study, Yadav and co-workers found important hysteresis in contact angle after exposing pAA brushes to solutions of different pH, obtaining different contact angle results upon increasing or decreasing pH. They correlated the observed hysteresis with the dispersity of the brushes.⁴⁵ In a recent QCM-D study, Hollingsworth and Larson reported significant hysteresis in swelling/shrinking pH cycles for pAA brushes and for sufficiently large molecular weight and not very large grafting density, although only a limited number of pH conditions were explored in this work.³⁰ Hysteretic response to pH changes has also been reported for other weak polyelectrolytes. For instance, Willot and co-workers reported important hysteresis in the swelling of a brush made of the polybase pDEA in *in situ* ellipsometry experiments.⁶⁸ We believe that these observations are related to the memory effects observed in this work. Applying an external potential drives the system to a significantly different state, which may be temporarily frozen after the field is removed (for instance, by formation of interchain hydrogen bonding for pAA, by hydrophobic interaction for the case of pDMAEMA, or by polymer–substrate dispersive forces). Our results suggest that this field-driven (meta) stable state can be long lasting, requiring a larger back potential step to achieve a complete reversal to the original equilibrium condition.

CONCLUSIONS

The body of results described in this work illustrates how weak PE brushes respond to the application of an external electric potential. Different driving forces, of chemical or physical origin,

can be engineered to efficiently manipulate the state of the brushes. The most remarkable changes occur when the pH of the solution is close to the pK_a of the grafted brushes, and significant conformational changes can be triggered. Although often out-of-equilibrium and not long lasting, these large electroconformational changes, not realizable with strong polyelectrolytes, can find applications in dynamic gating (e.g., opening/closing of channels²²), material transport (e.g., via local control of wettability properties^{69,18}), or temporal control of surface properties (e.g., adhesion, wettability, and fouling). For fully ionized brushes, less significant changes occurred when the surface potential was varied. These changes are determined by the electrostatic interaction between the charged wall and the polyanion, mediated by other ions in the environment. Even though the magnitude of the conformational changes is smaller in this case, they can find interesting applications for tuning static and dynamic surface properties in smart materials (e.g., lubrication, wettability, fouling, and adhesion), as we have shown in the past.¹⁹ A complete understanding of electrostimulation of polymer brushes will require to extend its theoretical description, often limited to thermodynamic equilibrium, to out-of-equilibrium, history-dependent conditions.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.macromol.1c02377>.

Complementary data: response of a bare gold electrode to an applied potential difference (QCM), relaxation of the open circuit potential (OCP) of polymer-coated electrodes after application of a potential step, and QCM electroresponse of polymer grafted on a partially blocked electrode (PDF)

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<https://pubs.acs.org/doi/10.1021/acs.macromol.1c02377>

Notes

The authors declare no competing financial interest.

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